Lab 7: Fuzzing for input Validation Bugs

Name: Vikas Vittal Aihole
SRN: PES1UG24CS534
Section: I

1. **Original program run:** the unfixed salon_booking.py runs on valid inputs and returns Payable Bill: 844.0.

```
PS C:\Users\Vikas\OneDrive\Desktop\Sem_05\SE\SE_Microservice_Catal
g\lab7\before> python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut:45
Service and Duration: ('Haircut', 45.0)

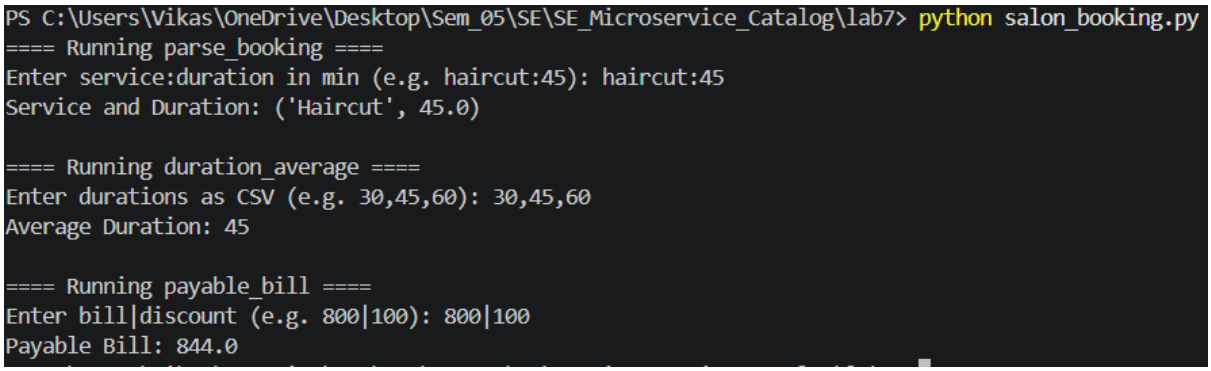
==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 30,45,60
Average Duration: 45

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): 800|100
Payable Bill: 844.0
```

2. **Original tests fail:** all 3 Hypothesis tests fail on None input with Attribute Error.

```
test_salon_booking.py:32: AssertionError
=================== short test summary info ====================
FAILED test_salon_booking.py::test_parse_booking_no_crash - AssertionError: parse_booking crashed with: AttributeError: 'N...
FAILED test_salon_booking.py::test_duration_average_no_crash - AssertionError: duration_average crashed with: AttributeError:...
FAILED test_salon_booking.py::test_payable_bill_no_crash - AssertionError: payable_bill crashed with: AttributeError: 'No...
====================== 3 failed in 1.34s =======================
```

3. **Fixed program run:** the corrected salon_booking.py gives the same results on valid inputs.



4. **Fixed tests pass:** all 3 Hypothesis tests pass after the input validation fixes.

